

Work Sample Context

Audience: Elementary students

Context: Originally designed and taught during a science education program in Kenya and later adapted for an elementary classroom in California.

Learning approach: Designed to make an abstract biological process tangible through movement and active learning. Students use a dance to model circulation, measure changes in their own heart rates, and physically move through a classroom-sized model of the circulatory system while carrying oxygen and carbon dioxide.

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Lesson Title: Heart and Blood Flow

Description: This lesson will allow students to learn the major parts of the heart and the general function and flow of the circulatory system. They will move around and engage in active learning through a simple dance and by travelling through a model of the circulatory system. This lesson can be modified for more advanced classes by adding more vocabulary (such as capillaries, aorta, and pulmonary arteries) and emphasizing the transfer of oxygen.

Level of class: Beginner

Learning objectives:

1. Students will be able to identify the following parts of the heart/circulatory system and understand how they work: atria, ventricles, veins, arteries, valves
2. Students will learn generally how the blood flows through the body and what makes up one heartbeat

Main Points:

1. Blood flows out of the heart through arteries and into the heart through veins
2. Blood flows in a cycle: body, right atrium, right ventricle, lungs, left atrium, left ventricle, body, etc.
3. A heartbeat is caused by muscles contracting/relaxing and valves opening/closing

Lesson:

1. Ask the students if they know what a heart looks like, have one student draw it. Explain that a typical heart shape is not actually the shape of the heart in your body. Show the poster of the labeled heart diagram (or drawing if the poster is not available). Ask students what they know about the heart or what they see.
2. Explain the major parts of the heart, write main vocabulary on the board.
 - Septum- large muscular “wall” separating the right side from the left side
 - Atrium/Atria- the top chambers of the heart
 - Ventricle/Ventricles- the bottom chambers of the heart
 - Valves- muscular flaps that work like trapdoors to keep blood flowing one way (the sound of your heartbeat comes from these valves closing)
 - Blood vessels- the tubes that carry blood to/from the heart
 - The heart is a muscle that pumps blood through your body. It beats by contracting and relaxing the muscles around each chamber.
3. Ask the students if they know where their hearts are. Have them point where they think it is on their own bodies. Show the students that the heart is in the middle of your body, in your chest, and it tilts toward the left. Have the students feel their hearts beating and count how many times it beats in 30 seconds. Explain that this is their heart rate.
4. Explain how blood flows in 2 circuits: from the lungs to the left side of the heart to the body and from the body to the right side of the heart to the lungs. Have them do the Circuit Dance (Activity 1). Ask what numbers they counted, then ask if these were higher or lower than

their resting heart rates. Explain that the heart delivers oxygen to the muscles that they use to jump around and the muscles need oxygen to move.

5. Explain that blood flows in through thin blood vessels called veins and out through thicker blood vessels called arteries. Have the students do Deep Breathing (Activity 2). Ask if their heart rates were lower or higher. Explain that when your muscles are relaxed, the heart does not have to work as hard.
6. How would everyone like to act like blood and see how it travels in the body? Have the students help you move the desks/chairs and set up the circulatory system model (Activity 3). Have them go through the circulatory system a few times then ask them questions about what they learned.

Activities:

1. Circuit Dance: To remember how the blood flows, we can use a simple dance—outstretch your arms to represent blood going to your lungs, then jump to the left, then wave your arms up and down to represent blood going from your head to your toes, then jump to the right. While doing this say “to the lungs to the left to the body to the right” then gradually get faster saying “lungs, left, body, right” while the students are all jumping around. After a few minutes, give everyone 30 seconds to check their heart rate again.
2. Deep Breathing: Have the students breathe deeply to slow down their heart rate while saying “In through the veins, out through the arteries.” Make sure they are relaxing and just focusing on slow, deep breaths. After a while, have everyone take their heart rate again for 30 seconds while continuing to breathe deeply.
3. Circulatory System Model: Set up the classroom as below (Figure 1). If there is enough yarn, put two strands for the “arteries” to represent that they are thicker.
 - a. Start with 2 students, one in each atrium. When they move to the ventricle, have another 2 students enter the atrium. Tell them that one heart beat is when the heart pumps blood in to the atrium and out through the ventricle, then it beats again. Continue until all students are in the circulatory system and are circling around the stations. When they get to a chair/desk, have them pick up a paper that says oxygen (at the lungs) or carbon dioxide (at the body) and drop off a paper that says carbon dioxide (at the lungs) or oxygen (at the body).

Materials:

1. Poster board with heart diagram, or draw simple heart with 4 chambers labeled
2. Posters for floor (can just be written on copy paper or whatever is available), written: “Right Atrium,” “Right Ventricle,” “Left Atrium,” “Left Ventricle”
3. Signs to put on chair or desk saying “LUNGS: Pick up Oxygen and Drop off Carbon Dioxide” and “BODY: Drop off Oxygen and Pick up Carbon Dioxide”
4. Small circles or squares of paper that say “O₂” (red) and “CO₂” (blue)
5. Yarn in 2 colors: blue and red, and scissors or a knife to cut
6. Duct tape or other strong tape

Figure 1: Circulatory System Set-Up

